Spam / Phishing Email Detection using Machine Learning

Artificial Intelligence Capstone Project

Course: Artificial Intelligence
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Abstract

Electronic mail has become one of the most important communication channels in modern society. However, the increasing volume of spam and phishing emails represents a significant challenge for users and organizations. These messages may contain misleading advertisements, malicious links, or attempts to steal sensitive information.

The objective of this project is to develop a machine learning system capable of automatically classifying emails as either legitimate (ham) or spam. The project follows a complete artificial intelligence pipeline including data exploration, preprocessing, feature extraction, model training, evaluation, and prediction. Two classical machine learning algorithms, Multinomial Naive Bayes and Logistic Regression, were implemented and compared using standard classification metrics.

The results demonstrate that machine learning techniques can effectively identify suspicious emails while remaining simple enough for a beginner-level artificial intelligence project.

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1. Introduction

Email remains one of the most widely used methods of communication in academic, personal, and professional environments. Unfortunately, the popularity of email has also made it a common target for spam and phishing attacks.

Spam emails often contain unwanted advertisements, promotional content, or misleading information. Phishing emails are designed to deceive users into revealing sensitive information such as passwords, banking details, or personal data.

Due to the large volume of emails received daily, manually reviewing each message is inefficient. Artificial intelligence and machine learning provide effective tools for automating this process. By learning patterns from previously labeled emails, machine learning models can predict whether a new email is likely to be spam or legitimate.

This project aims to demonstrate a complete AI workflow using simple and interpretable machine learning techniques suitable for a first university-level artificial intelligence course.

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2. Problem Statement

The primary objective of this project is to automatically classify emails into two categories:

- Ham (0): legitimate email messages.
- Spam (1): unwanted or potentially malicious messages.

The challenge is to build a model capable of learning meaningful patterns from email text and applying this knowledge to classify unseen messages accurately.

A successful solution should maximize the detection of spam emails while minimizing the number of legitimate emails incorrectly classified as spam.

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3. Dataset Description

The dataset used in this project is stored in the file `emails.csv`.

The dataset contains two attributes:

Column	Description
text	Email content
spam	Target label (0 = Ham, 1 = Spam)

The dataset contains labeled examples of both legitimate and spam emails, making it suitable for supervised machine learning classification.

Before training the models, the dataset was inspected to identify missing values, understand class distribution, and explore general characteristics of the email texts.

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4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to better understand the dataset before model training.

The following analyses were conducted:

4.1 Dataset Inspection

- Dataset dimensions were examined.
- Missing values were identified.
- Sample records were displayed.

4.2 Class Distribution Analysis

A count plot was generated to visualize the number of ham and spam emails.

This analysis helps determine whether the dataset is balanced or imbalanced. Significant class imbalance may negatively affect model performance.

```
Class distribution:
spam
0    4360
1    1368
Name: count, dtype: int64
```

Figure 1. Distribution of Ham and Spam Emails in the Dataset.

4.3 Email Length Analysis

The length of each email was calculated and visualized using a histogram.

The objective was to determine whether spam and ham messages exhibit different length characteristics.

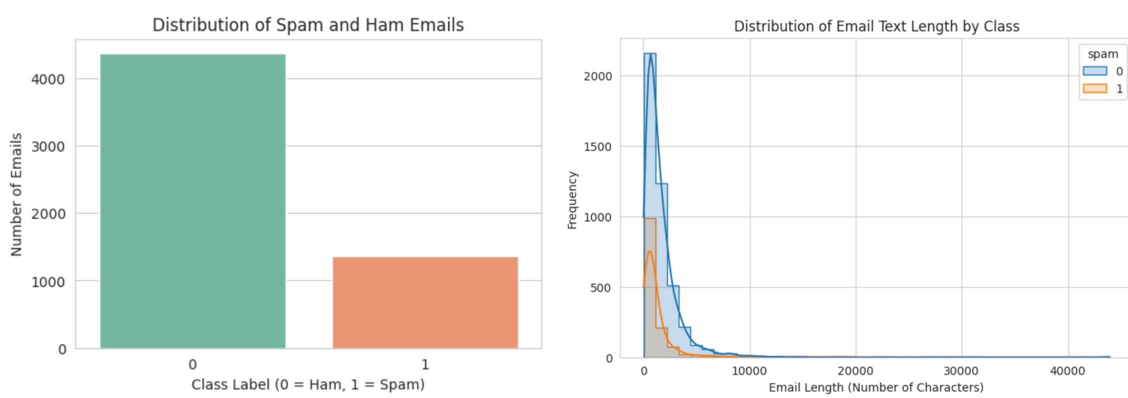

Figure 2. Distribution of Email Length by Class.

4.4 Common Word Analysis

The most frequent words appearing throughout the dataset were extracted and visualized using a bar chart.

This provided insight into the vocabulary commonly present in the email collection.

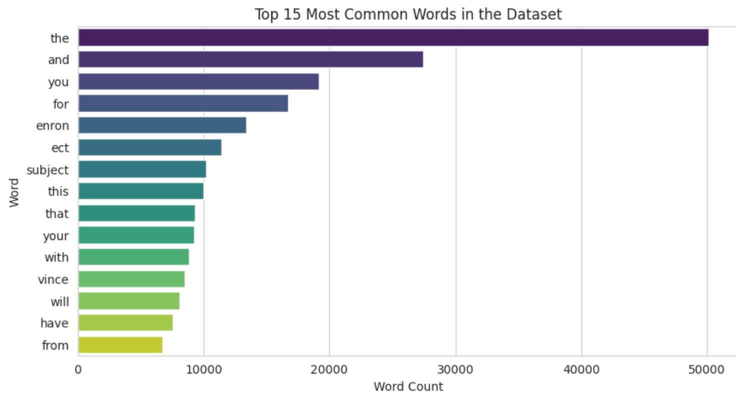

Figure 3. Top 15 Most Frequent Words in the Dataset.

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5. Data Preprocessing

Raw textual data often contains inconsistencies that may negatively impact machine learning performance.

Several preprocessing operations were applied:

5.1 Missing Value Handling

Missing email content was replaced with empty strings to ensure that all records could be processed successfully.

5.2 Text Normalization

The following transformations were applied:

- Conversion to lowercase
- Removal of excessive whitespace
- Basic text cleaning

These operations improve consistency and reduce unnecessary variations within the dataset.

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6. Feature Extraction

Machine learning algorithms require numerical input and cannot directly process raw text.

To convert email content into numerical features, the Term Frequency–Inverse Document Frequency (TF-IDF) technique was used.

TF-IDF measures the importance of words within a document relative to the entire dataset.

Advantages of TF-IDF include:

- Simple implementation
- Efficient computation
- Strong performance on text classification tasks
- Good interpretability

A maximum of 5000 features was used in the vectorization process.

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7. Model Development

Two machine learning models were implemented and compared.

7.1 Multinomial Naive Bayes

Multinomial Naive Bayes is widely used in text classification applications due to its simplicity and effectiveness.

Advantages:

- Fast training time
- Efficient memory usage
- Strong baseline performance for textual data

7.2 Logistic Regression

Logistic Regression is another popular classification algorithm that estimates class probabilities and provides robust performance on many machine learning tasks.

Advantages:

- Strong predictive performance
- Interpretable results
- Effective for binary classification problems

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8. Model Evaluation

To evaluate model performance, the dataset was divided into:

- 80% Training Data
- 20% Testing Data

Several evaluation metrics were calculated.

Accuracy

Measures the proportion of correctly classified emails.

Precision

Measures the percentage of emails predicted as spam that are actually spam.

Recall

Measures the percentage of actual spam emails successfully detected.

F1-Score

Provides a balance between precision and recall.

Confusion Matrix

A confusion matrix was generated for each model to visualize correct and incorrect predictions.

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9. Results and Discussion

	Model	Accuracy	Precision	Recall	F1-Score
0	Multinomial Naive Bayes	0.980803	0.973684	0.945255	0.959259
1	Logistic Regression	0.983421	0.984791	0.945255	0.964618

Table 1. Performance Comparison Between Multinomial Naive Bayes and Logistic Regression.

The evaluation results demonstrate the effectiveness of classical machine learning techniques for spam detection.

Both Multinomial Naive Bayes and Logistic Regression successfully learned meaningful patterns from the email texts after TF-IDF feature extraction.

The comparison of multiple metrics allows a more complete assessment than relying solely on accuracy.

The confusion matrices showed how each model handled spam and ham classifications and helped identify potential classification errors.

Overall, both models produced satisfactory results for a beginner-level artificial intelligence project.

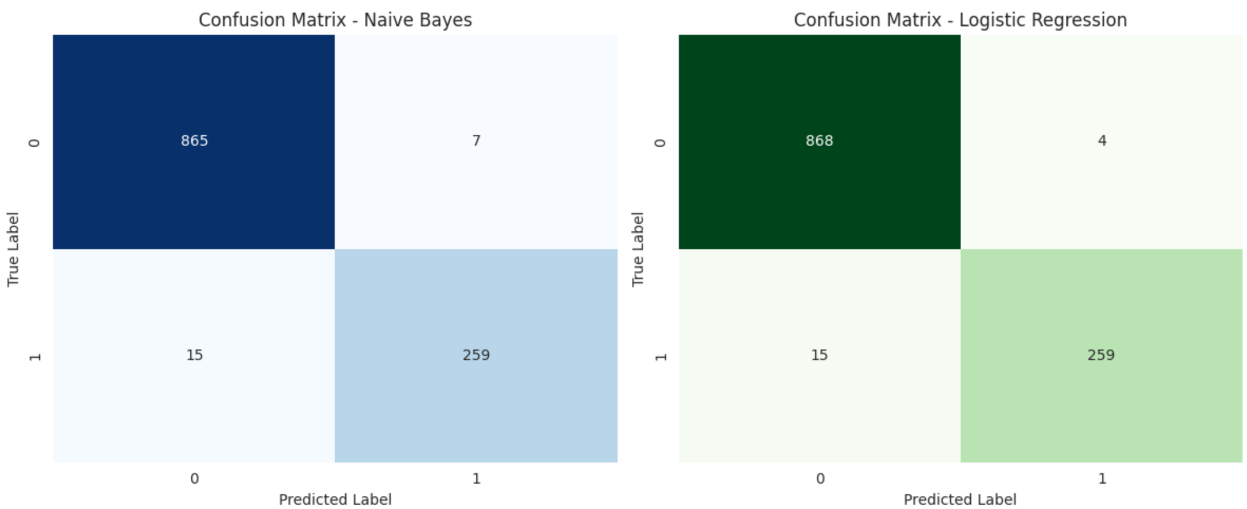

Figure 4. Confusion Matrix for Multinomial Naive Bayes and Logistic Regression..

10. Custom Email Prediction

A demonstration component was included in the notebook to allow prediction on custom email messages.

The user enters a new email message, and the trained model performs the following steps:

1. Text cleaning
2. TF-IDF transformation
3. Classification
4. Display of predicted label and probabilities

This feature demonstrates how the trained model can be applied to real-world inputs.

Examples:

```
Custom email text:


English Test Answers – Mohamed Reda El Ghazouani

Prediction: Ham
Probability [Ham, Spam]: [0.72115092 0.27884908]
```

Figure 5. Example of Custom Email Classification (Ham).

```
Custom email text:


Congratulations! You have won a free gift card. Click the link now to claim your prize.

Prediction: Spam
Probability [Ham, Spam]: [0.15144632 0.84855368]
```

Figure 6. Example of Custom Email Classification (Spam).

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11. Project Limitations

Although the project achieved its objectives, several limitations remain.

Dataset Dependency

The model's performance depends heavily on the quality and diversity of the dataset.

False Positives

Some legitimate emails may contain language patterns similar to spam messages, leading to incorrect classifications.

Limited Scope

The project intentionally uses simple preprocessing and traditional machine learning methods suitable for a first AI course.

Future Improvements

Potential improvements include:

- Larger and more diverse datasets
- Advanced text preprocessing
- N-gram feature extraction
- Hyperparameter optimization
- Deep learning approaches
- Specialized phishing detection features

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12. Conclusion

This project presented the development of a complete machine learning solution for spam and phishing email detection.

The work followed a structured artificial intelligence pipeline including data exploration, preprocessing, feature extraction, model training, evaluation, and prediction.

Using TF-IDF feature extraction combined with Multinomial Naive Bayes and Logistic Regression, the system was able to classify emails into spam and ham categories effectively.

Beyond the classification results, this project provided valuable experience with the practical stages of machine learning development and demonstrated how artificial intelligence can be applied to solve real-world problems.

The project successfully fulfills the objectives of the Artificial Intelligence Capstone Project while remaining accessible, understandable, and appropriate for a beginner-level university course.